



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431696.

Department of Civil Engineering

Summer Term Examination

Academic Year: 2024-25

Class: T.Y. B. Tech.

Course: Water Resources Engg.

Course Code: PCC-CE312

Date: 21.06.2025

Time: 10 a. m. to 01:00 p.m.

Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Name of Question	Marks	CO	BT
Q.1	a Explain in brief about following (i) Field capacity (ii) Permanent wilting point (iii) Consumptive use (iv) Effective rainfall	10	CO1	BT1
	b After how many days will you supply irrigation water to soil, if (i) Field capacity=28% (ii) Permanent wilting point=13% (iii) Density of soil=1.3 gm/cc (iv) Root zone depth=70 cm (v) Consumptive use =12 mm.	10	CO1	BT4
Q.2	a Enlist and explain the steps in a hydraulic design of a typical border irrigation system	10	CO2	BT2
	b Design a regime channel for a discharge of 50 cumecs and silt factor 1.1 using Lacey's theory.	10	CO2	BT4
Q.3	a Distinguish between weir and barrage using neat sketches.	10	CO3	BT2
	b Discuss various cross-drainage works with the help of neat sketches.	10	CO3	BT3
Q.4	a Discuss in brief about hydraulic, seepage and structural failures of earth dam.	10	CO4	BT2
	b Explain the line of seepage or phreatic line. Also explain the determination of phreatic line for homogeneous dam section with horizontal filter.	10	CO4	BT3
Q.5	a Explain the following forces acting on a gravity dam (i) Water pressure (ii) Uplift pressure. Derive the expressions for principal and shear stresses acting on a gravity dam.	10	CO5	BT2
	b What do you mean by elementary profile of a gravity dam? With the help of neat sketch explain about Ogee spillway	10	CO5	BT4

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

On successful completion of this course, students should be able to:			
CO 1	Determine irrigation requirement and schedule.	CO 2	Design surface irrigation systems.
CO 3	Design canal head works and cross-drainage works.	CO 4	Design earth dam.
CO 5	Design gravity dam and spillway.		



Shri Guru Gobind Singhji Institute of Engineering & Technology, Vishnupuri, Nanded-431606.
Department of Civil Engineering
Summer Term Examination
Academic Year: 2024-25

Class: TY B.Tech. Civil Engineering

Course: Elective-I: Prestressed Concrete

Course Code: PEC-CE 306

Date: 23/06/2025

Time: 10:00 pm to 13:00 pm

Max. Marks: 100

- Note**
- (i) All questions are compulsory;
 - (ii) Use of non-programmable calculator is permitted.
 - (iii) The symbols carry usual meaning.
 - (iv) Assume suitable data if necessary.
 - (v) This paper contains 02 page.

Q. No.	Question	Marks	CO	BT
1 (a)	Write short note on types of prestressing steel and explain the necessity of use high tensile strength steel in prestressing.	04	CO1	BT2
1 (b)	Explain any four losses in pre-stressed member.	04	CO1	BT2
1 (c)	Explain P-line and C-line with neat sketches.	04	CO2	BT2
1 (d)	Explain the concept of prestressing a structural element. Also explain pre-tensioning and post tensioning methods with neat sketches.	06	CO1	BT2
2 (a)	A prestressed concrete beam is post-tensioned by a cable carrying an initial stress of 1150 N/mm^2 . The slip at the jacking end was observed to be 3 mm. The modulus of elasticity of the concrete is 200 GPa. Estimate the percentage loss of stress due to anchorage slip if the length of the beam is (i) 5.4 m and (ii) 50 m.	04	CO3	BT3 BT4
2 (b)	A post tensioned prestressed beam of span-length 12 m has a rectangular section 350 mm wide and 900 mm deep. The beam is prestressed by two cables. The first cable is parabolic with an eccentricity of 280 mm below the centroidal axis at the center of the span and zero at the support. The second cable has a uniform eccentricity of 280 mm below the centroidal axis. The area of each cable is 100 mm^2 and the initial stress in each cable is 1150 N/mm^2 . Calculate the loss of stress in these cables if the cables are tensioned (i) simultaneously and (ii) successively.	08	CO3	BT3 BT4
2 (c)	A prestressed concrete beam of section 150 mm wide and 350 mm deep is used over an effective span of 7 m to support a uniformly distributed load of 4 kN/m , which includes the self-weight of the beam. The beam is prestressed by a straight cable carrying a force of 200 kN and located at an eccentricity of 50 mm. Determine the location of the thrust line at support, quarter span, and central span section and plot its position.	08	CO2	BT3 BT4
2 (d)	A rectangular concrete beam of cross section 180 mm wide and 350 mm deep is prestressed by a straight cable carrying an effective force of 200 kN at an eccentricity of 70 mm. The beam supports an imposed load of 4 kN/m over a span of 7 m. If the modulus of rupture of the concrete is 5.2 N/mm^2 , evaluate the load factor against cracking assuming the self weight of concrete as 24 kN/m^3 .	04	CO2	BT3 BT4
3 (a)	A rectangular concrete beam 120 mm wide by 300 mm deep spanning over 8 m is prestressed by a straight cable carrying an effective prestressing force of 250 kN located at an eccentricity of 40 mm. The beam supports a live load of 1.2 kN/m . i) Calculate the resultant stress distribution for the center-of-span cross section of the beam assuming the density of concrete as 24 kN/m^3 .	10	CO2	BT4

Q. No.	Question	Marks	CO	BT
	ii) Find the magnitude of the prestressing force with an eccentricity of 40 mm which can balance the stresses due to dead load and live load at the soffit of the center of span section.			
3 (b)	A concrete beam of 10 m span, 120 mm wide and 300 mm deep, is prestressed by two cables. The area of each cable is 280 mm ² and the initial stress in each cable is 1150 N/mm ² . Cable 1 is parabolic with eccentricity of 65 above the centroid at the support and 65 mm below the centroid at the mid span. Cable 2 is straight with uniform eccentricity of 65 mm below the centroidal axis of the beam. If the cables are tensioned from one end only, estimate the percentage loss of stress in each cable due to friction. Assume $\mu = 0.32$ and $k = 0.0015$ per meter.	08	CO3	BT4
3 (c)	A prestressed concrete beam is pre-tensioned with a prestressing cable with an initial stress of 1280 N/mm ² . The ultimate tensile strength of cable is 1600 N/mm ² . Calculate the loss of stress in cable due to relaxation of stress in steel as per the norm in IS 1343 for: (a) Normal Relaxation, and (b) Low relaxation.	04	CO3	BT3
3 (d)	A post tensioned prestressed rectangular beam of span 12 m is subjected to an imposed uniformly distributed load of 8 kN/m. The loss of prestress is 11 percent. Calculate the minimum prestressing force required and the corresponding eccentricity of the prestressing force required. Take, (i) The width of the beam = 320 mm. (ii) Depth of the beam = 680 mm Allowable tensile stress = $f_{tt} = f_{tw} = 0 \text{ N/mm}^2$	08	CO2	BT4
4 (a)	A post tensioned prestressed rectangular beam of span 10 m is subjected to an imposed uniformly distributed load of 11 kN/m. the stresses in concrete should not exceed 12 N/mm ² in compression and 1.6 N/mm ² in tension. The loss of prestress is 15 percent. Calculate the minimum possible depth. The width of the beam is to be taken as 400 mm.	06	CO2	BT4
4 (b)	A pretensioned T section has a flange 1400 mm wide and 175 mm thick. The width and the depth of the rib are 325 mm and 1800 mm respectively. The high-tensile steel has an area of 5400 mm ² and is located at an effective depth of 1800 mm. If the characteristic cube strength of the concrete and the tensile strength of steel are 40 N/mm ² and 1600 N/mm ² respectively, evaluate the flexural strength of the T section	12	CO2	BT4
4 (c)	A rectangular concrete beam 300 mm wide and 800 mm deep supports two concentrated loads of 20 kN each at the third point of span 9 m. (i) Suggest a suitable cable profile (ii) If the eccentricity of the cable profile is 100 mm for the middle third portion of the beam, calculate the prestressing force required to balance the bending effect of the concentrated loads (neglect self-weight of the beam). (iii) For the same cable profile, find the effective force in the cable if the resultant stress due to self-weight, imposed loads and prestressing force is zero at the bottom fibre of the mid-span section. Assume unit weight of concrete to be 24 kN/m ³ .	10	CO2	BT3

Course Outcome (CO)		Bloom's Taxonomy (BT)
CO 1	Understand the basic concepts of Prestressed concrete and identify the materials for prestressing.	BT1 - Remember
CO 2	Analyse and design components of Prestressed concrete section for flexure and shear.	BT2 - Understand
CO 3	Estimate Losses of prestressing and apply to real life problems while designing PSC members.	BT3 - Apply
CO 4	Evaluate the performance of Prestressed concrete beams for shear and bond.	BT4 - Analyze
		BT5 - Evaluate
		BT6 - Create

--- ALL THE BEST ---



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
Summer Term Examination
Academic Year: 2024-25 (Semester – I)

Class: Third Year (T.Y) B.Tech.	Subject: Transportation Engineering	Subject Code: PCC-CE304
Date: 20.06.2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- i. *Mobile phones and programmable calculators are strictly prohibited.*
- ii. *Figures to the right indicate full marks.*

Q.N.	Questions	Marks	CO	BT
Q.1	Explain in detail the ductility test of bitumen with neat sketch.	05	1	2
Q.2	Explain the Following: 1) Difference between Macadam and Telford method. 2) PIEV theory in details with neat sketch.	10	1,3	2
Q.3	Explain the Following: 1) Various methods of carrying out speed and delay study. 2) On-street and Off-street parking in details with neat sketch.	20	3	2
Q.4	Explain the Following: 1) Types of rails in details with neat sketch. 2) Necessity of Intelligent Transportation system.	20	4,5	2
Q.5	A valley curve has a descending gradient of 1 in 20 followed by an ascending gradient of 1 in 35. Find the length of the valley curve required for a design speed of 60 km/hr for comfort condition.	05	2	3
Q.6	Calculate equivalent single wheel load of a dual wheel assembly 1400 kg each for pavement thickness 270 mm centre to centre tyre spacing is 300 mm and distance between the walls of the tyres is 80 mm.	05	2	3

Q.7	The laboratory data for a soil is given below. Using these values find the group index value. <table><tr><td>Sieve</td><td>10</td><td>40</td><td>200</td><td>Liquid limit (%)</td><td>Plastic limit (%)</td><td>Plasticity Index</td></tr><tr><td>% passing</td><td>96</td><td>89</td><td>70</td><td>49</td><td>24</td><td>25</td></tr></table>	Sieve	10	40	200	Liquid limit (%)	Plastic limit (%)	Plasticity Index	% passing	96	89	70	49	24	25	05	1	3
Sieve	10	40	200	Liquid limit (%)	Plastic limit (%)	Plasticity Index												
% passing	96	89	70	49	24	25												
Q.8	The area of a certain district in India is 80,000 sq. km and there are 86 towns as per 1981 census. Determine the lengths of NH, SH, MDR, ODR and VR along with total length of road to be provided in this district by the year 2001 using third year road development plan.	10	1	3														
Q.9	While aligning a highway in a built-up area, it was necessary to provide a horizontal circular curve of radius 400 m. The design speed is 80 km/hr, length of wheelbase of largest truck is 6 m and width of pavement is 7 m. Design the following geometric features: - (a) Superelevation (b) Extra widening of pavement (c) Length of transition curve as per allowable rate of centrifugal acceleration and as per IRC on plain and rolling terrain. Take centre line rotation 1 in 100.	10	2	3														
Q.10	Calculate the maximum speed of a train on a Broad gauge (B.G) track having a curvature of 5° and a cant of 80 mm. Assume the cant deficiency as 7.6 cm and value of G is 1.676 m for B.G and find out the length of the curve for a B.G curved track if maximum permissible speed on curve is 85 km/hr.	10	4	3														

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO 1	Comprehend the concepts of road development, road alignment and preparation of highway project.
CO 2	Design cross section elements, sight distance, horizontal and vertical alignment. Study, analysis and design of curves and grades.
CO 3	Implement traffic studies, traffic regulations and control, and intersection design
CO 4	Design various components of railway engineering
CO 5	Understand the fundamentals of urban transportation systems



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
Summer Term Examination (STE)
Academic Year: 2024-25 (Semester – I)

Class: Third Year
Date: 19-06-2025

Subject: Environmental Engineering
Time: 10:00-13:00

Subject Code: PCC-CE-302
Maximum Marks: 100

Note:

- i. Assume appropriate data and state assumption clearly.
- ii. Non-programmable calculator is allowed.
- iii. Exchange or sharing of any material is not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Question statement	Marks	CO	BT										
Q.1A	Draw a typical layout of municipal water treatment plant for treatment of river sourced raw water.	05	CO2	BT3										
Q.1B	Explain screen chamber types and function in wastewater treatment plant.	05	CO4	BT3										
Q.1C	Explain back washing of rapid sand filter.	05	CO2	BT3										
Q.1D	Explain break point chlorination with neat figure	05	CO2	BT3										
Q.2a	Which Indian standard code laid down the water quality desirable and permissible limits for drinking water. Write the desirable values or range for following based on IS code for drinking water. a. pH, b. Hardness, c. Alkalinity, and d. MPN	10	CO1	BT1										
Q.2b	A river due to excess sewage discharge turned into nala. The BOD ₅ measured at some point was 186 mg/L. It was observed that the rate of deoxygenation was 0.17/day (base e) at 20°C. The river flowing through the locality at average temperature of 30.9°C. How many days will it take to reduce BOD of river (nala) water to 25 mg/L.	10	CO4	BT3, BT4, BT5										
Q.3A	Define sewage. Enumerate and explain in brief sources of wastewater in sewers.	05	CO4	BT1, BT2										
Q.3B	Enumerate the technologies available for disposal of solid waste. Explain sanitary landfill with neat sketch.	05	CO5	BT3										
Q.3C	Estimate population using arithmetic increase method for a town having population as specified below for supplying water for 15 years. The plant commences its operation in year 2025. <table border="1" style="margin: 10px auto; width: 80%;"> <tr> <td>Year</td><td>1991</td><td>2001</td><td>2011</td><td>2021</td></tr> <tr> <td>Population</td><td>35000</td><td>49000</td><td>67000</td><td>80000</td></tr> </table>	Year	1991	2001	2011	2021	Population	35000	49000	67000	80000	05	CO1	BT4
Year	1991	2001	2011	2021										
Population	35000	49000	67000	80000										
Q.3D	Determine the discharge of wastewater through a sewer running full. Sewer has diameter of 35 cm and laid at a gradient of 1 in 120. Adopt manning's coefficient of 0.015.	05	CO3	BT4										
Q.4A	Design a standard rate trickling filter for a town with population of 50 thousand with water supply rate of 225 LPCD. The average BOD of domestic wastewater observed as 270 mg/L and about 25% BOD is removed in primary treatment. The effluent BOD shall not be more than 18 mg/L. Use NRC equation.	10	CO4, CO5	BT5										
Q.4B	The wastewater inflow to an activated sludge plant (ASP) is 5000 L/s, and the plant is to be operated with a food to microorganism ratio of	10	CO5	BT5										

Q.N.	Question statement	Marks	CO	BT
	0.2 mg/mg-d. The concentration of influent biodegradable organic matter of the wastewater to the plant (after primary settling) is 140 mg/L, and the mixed liquor volatile suspended solids (MLVSS) concentration to be maintained in the plant is 2000 mg/L. Assuming complete removal of biodegradable organic matter in the ASP, determine the volume required for tank.			
Q.5A	Write a note on oxidation ponds system incorporating classification of ponds, mechanism of purification and factors affecting pond systems.	10	CO4	BT3
Q.5B	Determine the quantity alum required in a month to treat 20 MLD water. The average monthly alum dose is 12 mg/L. Alum under sufficient alkalinity emanates carbon dioxide as presented in the reaction. Determine the CO ₂ emitted in atmosphere from clariflocculator of plant a month and year. $\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O} + \text{Ca}(\text{HCO}_3)_2 \rightarrow 2\text{Al}(\text{OH})_3\downarrow + 3\text{CaSO}_4 + 18\text{H}_2\text{O} + 6\text{CO}_2\uparrow$	10	CO3	BT4

***** ALL THE BEST *****

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

COs	On successful completion of this course, students should be able to:
CO1	Identify the source of water and water demand.
CO2	Understand the water treatment concept and methods.
CO3	Apply water distribution processes and operation and maintenance of water.
CO4	Analyze the sewage characteristics and design various sewage treatment plants.
CO5	Evaluate basic process designs of water and wastewater treatment plants collect, reduce, analyse, and evaluate basic water quality data.